

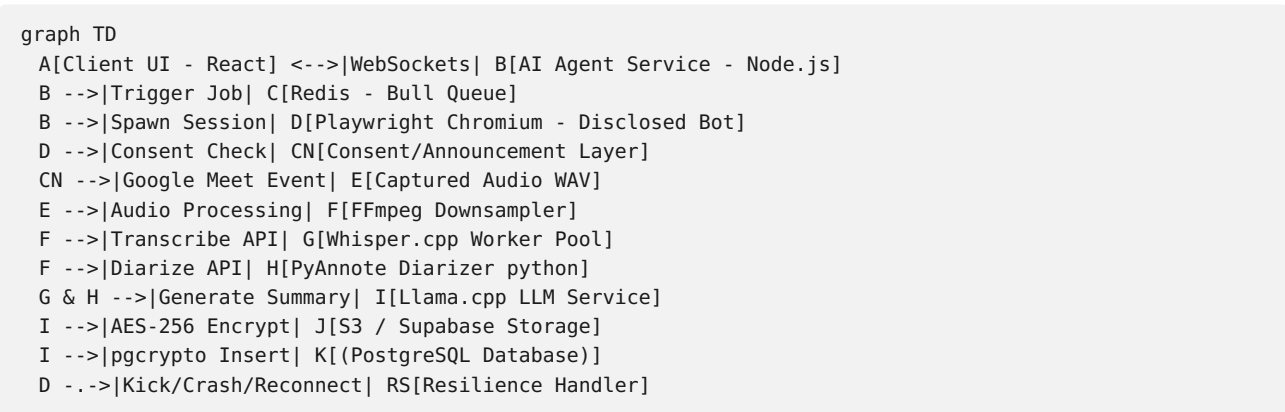
MeetIQ AI Meeting Agent - Implementation Plan v2

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Architect, implement, and verify a self-contained, disclosed **AI Meeting Agent** incorporating Playwright automation, WebSockets, Redis queues, Whisper.cpp/Llama.cpp self-hosted engines, AES-256 data encryption, and Kubernetes deployments.

Naming note: Renamed from “Ghost in the Meet” / “stealth automation.” The bot must be a visibly disclosed participant, not a covert one — this shapes both legal posture and engineering goals (compliant behavior vs. detection evasion).

System Topology & Microservices



Proposed Changes

1. Database Schema & Security (`schema.prisma`)

- **Add** `pgcrypto` support: column-level encryption for transcripts using sym-encryption.
- **Add** tracking columns for agent performance, calendar keys, and S3 encryption tags.
- **Add** `retention_policy` table: per-workspace/per-user configurable retention window (default 30 days, overridable), plus `consent_status` and `consent_log` fields per meeting.

2. Standalone Agent Service (`ai-agent-service/`)

File	Purpose
<code>meet-bot.js</code> [NEW]	Playwright Chromium automation with visible bot identity (labeled name/avatar), Google Account auth flow, lobby handling
<code>consent-announcer.js</code> [NEW]	Posts an in-meeting chat message / on-screen indicator when recording starts; supports host-configurable opt-out
<code>transcriber-whisper.js</code> [NEW]	Local execution handler for Whisper.cpp binaries, dispatched via worker pool
<code>summarizer-llama.js</code> [NEW]	Local execution handler for Llama.cpp GGML/GGUF models
<code>crypt-helper.js</code> [NEW]	AES-256-CBC file encryption/decryption utilities
<code>calendar-monitor.js</code> [NEW]	Google Calendar OAuth polling and event queuing, scoped to the requesting user’s own calendar/org permissions

<code>cron-retention.js</code> [NEW]	Cleans up files/DB records per the configurable <code>retention_policy</code> , not a hardcoded 30 days
<code>resilience-handler.js</code> [NEW]	Detects bot removal/kick, lobby timeout, or Playwright session crash; handles reconnect attempts and flushes partial audio before failing
<code>auth-gateway.js</code> [NEW]	Validates that a Client UI request to join a meeting is authorized against the calendar event owner/invitee list — prevents cross-user join triggering

3. Consent & Disclosure Layer *(new section)*

- Bot joins with a clearly labeled name (e.g. “MeetIQ Notetaker”) and avatar — never blank/anonymous.
- On join, posts an announcement in meeting chat and/or triggers a visible on-screen banner where the platform supports it.
- Meeting organizer can require explicit participant opt-in before recording begins (configurable per workspace, since consent law varies by jurisdiction — some are one-party, some two-party/all-party consent).
- If a participant objects, the bot supports a “leave/stop recording” command.
- All consent events are logged (`consent_log`) alongside the transcript for audit purposes.

4. Containerization & Orchestration Configs

- `Dockerfile.agent` **[NEW]**: Multi-stage Dockerfile with Playwright, Node, and FFmpeg.
- `Dockerfile.whisper` **[NEW]**: CUDA GPU-enabled container for Whisper.cpp.
- `Dockerfile.llama` **[NEW]**: GGML container for Llama.cpp.
- `docker-compose.staging.yml` **[NEW]**: Compose setup linking Agent, Redis, Whisper.cpp, and Llama.cpp.
- `kubernetes/` **[NEW]**: Manifests for deployments, services, ingress, and **HPA (Horizontal Pod Autoscaler)** for the Whisper/Llama worker pools, so GPU containers scale with queue depth instead of running one-per-session.

5. Documentation

- `RUNBOOK.md` **[NEW]**: Troubleshooting, manual join override, emergency shutdown, and kick/crash recovery procedures.
- `THEME.md` **[NEW]**: Style guidelines mapping components to Tactical Steel Slate & Ochre/Sage theme variables.
- `CONSENT_POLICY.md` **[NEW]**: Jurisdiction-aware guidance for workspace admins on configuring consent requirements.

Verification Plan

Automated Tests

- Jest suites verifying AES encryption, local transcription parsing, and calendar polling logic.
- New: tests for `auth-gateway.js` cross-user join prevention.
- New: tests for `resilience-handler.js` — simulate kick, crash, and lobby timeout scenarios.
- New: tests for `consent-announcer.js` — verify announcement fires before audio capture begins, and opt-out halts recording.

Bot-Join Flow Testing *(new)*

- Since Meet’s DOM and bot-detection heuristics change over time, run scheduled integration tests (not just at build time) against a real/staging Meet instance to catch breakage early.
- Test across meeting sizes (2, 10, 50+ participants) and degraded network conditions for audio capture quality.

Load & Security Verification

- Verify Docker build compiles with zero critical vulnerabilities.
- Run type checks and compiler tests on deployment targets.
- Load-test the Whisper/Llama worker pool under concurrent meeting sessions to validate autoscaling and cost bounds (GPU time is the main cost driver — this needs a number before launch, not after).

Compliance Review (*new*)

- Legal/compliance sign-off on default consent behavior per target region before enabling by default in any jurisdiction with all-party consent requirements.